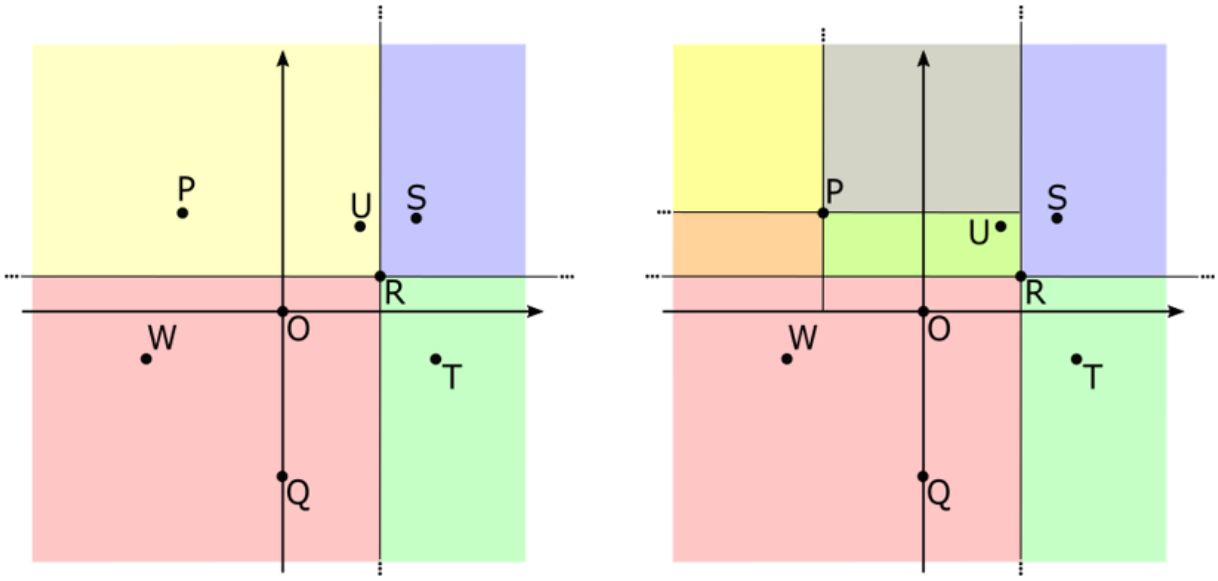


Figure 8.10 Four-way partitioning of the plane. For each point, we split the area it belongs to in 4 quadrants using the lines passing through the point and parallel to the axes. For the first one, `R` in the example, we get 4 infinite quadrants (left). When we choose the next point, `P` (right) we have to divide the yellow quadrant further in 4 parts: we get a finite rectangle, to which `U` belongs, and 3 infinite sections of the plane. For each further point, like `U`, we further split the region it belongs to in 4 more smaller regions.

Now the question is: can we extend this solution to $\mathbb{R}^3$? To move to a 3-D space, we need to answer one question: into how many sub-hyperplanes do we split the plane for each point? In 1-D, it was 2 segments for each point, in 2-D it was 4 quadrants, and similarly in 3-D we will need 8 octants[3].

Therefore, for each point in the dataset we will add a node with 8 children, one for each octant resulting from splitting